 # Physical Account -- 8-Class Resource Ledger & Errata Verification (1 Page)

1. Subcomponent Mass, Position & Stability Ledger (Organizer Errata Applied)
Stability is computed strictly **per subcomponent and assigned position** ($\times 1.0$ main deck bays ‘bay1‘/‘bay2‘, $\times 1.2$ ‘hangar‘/‘hangar_roof‘/‘flight_deck‘, $\times 3.0$ ‘mast_3‘), never by multiplying an entire multi-part guest by the mast factor. Local structural limits (‘Bay 1 <= 8,000 kg‘, ‘Bay 2 <= 4,000 kg‘, ‘Hangar <= 1,000 kg‘, ‘VTOL airframe <= 150 kg‘, ‘mast_3 <= 60 kg / 0.6 m2‘, ‘hangar_roof <= 150 kg / 2.0 m2‘) are enforced alongside global limits (‘10,000 kg‘ mass, ‘12,000 kg-eq‘ stability). Under the organizer errata, **G4 hangar equipment is 850 kg** (overriding PDF 900 kg) + **140 kg drone = 990 kg**, and **G5 includes both the 3,000 kg container and 25 kg antenna = 3,025 kg**:

Guest ID & Name	Subcomponents & Assigned Positions	Mass (kg)	Stability Multiplier & Calculation	Stability (kg-eq)	Electrical, Thermal & Data Envelope
G1 Radar/EO Meteo	‘G1_MAST_SENSOR_HEAD‘ $\rightarrow$ ‘mast_3‘ (‘0.5 m2‘)	‘45‘	45×3.0 (‘mast_3‘)	‘135.0‘	‘1.5 kW‘ air on ‘mast_3‘ ($\le 2\text{ kW}$ limit; ‘0 W‘ water); ‘50 Mbps‘ LAN, ‘200 kbps‘ SATCOM
G2 Ocean Winch/Probe	‘G2_ROV_WINCH_CONTAINER_10FT‘ $\rightarrow$ ‘bay2‘ / ‘bay1_a‘	‘3,500‘	$3,500 \times 1.0$ (deck bay)	‘3,500.0‘	‘20 kW‘ sustained ‘300 s‘ winch / ‘6 kW‘ continuous $\rightarrow$ water; ‘5 Mbps‘ LAN, ‘500 kbps‘ SATCOM
G3 AUV Survey/Cradle	‘G3_AUV_CRADLE_BAY1_20FT‘ $\rightarrow$ ‘bay1‘ (‘20 ft‘ via ramp)	‘5,500‘	$5,500 \times 1.0$ (deck bay)	‘5,500.0‘	‘1 kW‘ survey / ‘25 kW‘ (‘3,600 s‘) uninterruptible charge $\rightarrow$ water; ‘100 Mbps‘ cradle LAN
G4 VTOL SAR *(Errata)*	‘G4_DRONE_VEHICLE‘ (‘140 kg‘) + ‘G4_HANGAR_STATION‘ (‘850 kg‘) $\rightarrow$ ‘hangar‘	‘990‘	$(140 + 850) \times 1.2$ (‘hangar‘ $\le 1,000\text{ kg}$)	‘1,188.0‘	‘35 kW‘ (‘1,200 s‘) recovery/charge, ‘10 kW‘ patrol, ‘2 kW‘ standby (‘0 bps‘ silence); ‘20 Mbps‘ LAN
G5 Comms/Compute	‘G5_COMPUTE_CONTAINER‘ (‘3,000 kg‘ $\rightarrow$ ‘bay2‘/‘bay1_a‘) + ‘G5_MAST_ANTENNA‘ (‘25 kg‘ $\rightarrow$ ‘mast_3‘/‘hangar_roof‘)	‘3,025‘	$3,000 \times 1.0 + 25 \times 3.0$ (‘mast_3‘) or $+ 25 \times 1.2$ (‘roof‘)	‘3,075.0‘ (‘3,030.0‘ roof)	‘45 kW‘ (‘43.8 kW‘ water + ‘1.2 kW‘ air) / ‘8 kW‘ degraded (‘30%‘ retention); ‘1 Gbps‘ LAN, ‘6 Mbps‘ SATCOM
G6 Civil Hydro Lab	‘G6_HYDRO_LAB_10FT‘ $\rightarrow$ ‘bay1_a‘ / ‘bay2‘ *(All fields ‘ASSUMED‘)*	‘420‘	420×1.0 (deck bay)	‘420.0‘	‘4.5 kW‘ (‘300 s‘) $\rightarrow$ ‘4.5 kW‘ water; ‘25 Mbps‘ LAN, ‘250 kbps‘ SATCOM

2. Five Combinations of Four Official Guests (‘{G1..G5}‘): Mass vs. Full 8-Class Admission
Evaluating all $\binom{5}{4} = 5$ four-guest combinations confirms the organizer errata (**exact 3 of 5 exceed the 10,000 kg margin**) and proves why **passing mass does not imply passing placement, mast/roof clearance, or 14-day Mission Envelope A.10 constraints**:

4-Guest Set	Total Mass (kg)	$>10,000\text{ kg}$?	Total Stability (kg-eq)	Physical Placement, Mast/Roof & Mission Envelope Verdict (D1)
‘{G1, G2, G3, G4}‘	**10,035 kg**	**YES** (‘+35 kg‘)	‘10,323.0‘ ($\le 12,000$)	‘YES_IF‘ (Exceeds ‘10,000 kg‘ by ‘35 kg‘; displacing ‘G2‘ yields ‘6,535 kg‘)
‘{G1, G2, G3, G5}‘	**12,070 kg**	**YES** (‘+2,070 kg‘)	‘12,210.0‘ ($> 12,000$)	‘YES_IF‘ (Exceeds mass & stability; ‘G1‘ & ‘G5‘ collide on ‘mast_3‘ and ‘1.5+1.2=2.7 kW > 2.0 kW‘ mast air)
‘{G1, G2, G4, G5}‘	**7,560 kg**	**NO** (‘2,440 kg‘ free)	‘7,898.0‘ ($\le 12,000$)	**Blocked on ‘mast_3‘ / ‘hangar_roof‘**: ‘G5‘ antenna collides with ‘G1‘ on ‘mast_3‘ (‘2.7 kW > 2 kW‘ air) and with ‘G4‘ drone clearance on ‘hangar_roof‘
‘{G1, G3, G4, G5}‘	**9,560 kg**	**NO** (‘440 kg‘ free)	‘9,898.0‘ ($\le 12,000$)	**Blocked on ‘mast_3‘ / ‘hangar_roof‘**: passes mass/stability (‘G5‘ container fits in ‘bay2‘), but ‘G5‘ antenna collides with ‘G1‘ (‘mast_3‘) or ‘G4‘ (‘hangar_roof‘)
‘{G2, G3, G4, G5}‘	**13,015 kg**	**YES** (‘+3,015 kg‘)	‘13,263.0‘ ($> 12,000$)	‘YES_IF‘ (Exceeds mass & stability; ‘G2‘+‘G5‘ containers cannot share ‘bay2‘ (‘6,500 kg > 4,000 kg‘) while ‘G3‘ occupies ‘bay1‘)

*(Note: Replacing ‘G5‘ with custom card ‘G6‘ in ‘{G1, G2, G4, G6}‘ yields ‘4,955 kg‘, ‘5,243.0 kg-eq‘, ‘G1‘ on ‘mast_3‘, ‘G2‘ on ‘bay2‘, ‘G4‘ in ‘hangar‘, and ‘G6‘ in ‘bay1_a‘ $\rightarrow$ **‘YES‘ across all 8 resource classes**.)*

3. Cooling Curve, Thermal Inertia & Scenario B Power Accounting
- **Chilled-Water Curve**: $C(T) = 160,000 - 40,000 \frac{T-12}{18}\text{ W}$ for $T \in [12, 30]\,^\circ\text{C}$. At $30\,^\circ\text{C}$, $C(30) = 120\text{ kW}$ nominal; with -40% loop degradation, $C_{\text{deg}}(30) = 72\text{ kW}$. Sustained loads ($>60\text{ s}$) cannot borrow from the $+30\text{ kW}/60\text{ s}$ (‘1.8 MJ‘) thermal inertia buffer.
- **Scenario B Power vs. Cooling Binding**: During G4 recovery/charge (‘35 kW‘) + G3 quiet survey (‘1 kW‘) + G5 full spectrum (‘45 kW‘), instantaneous electrical draw is **$81\text{ kW} \le 100\text{ kW}$** (no instantaneous electrical deficit on generator loss), whereas **firm committed power** including G3’s uninterruptible ‘25 kW‘ turnaround recharge reservation is **$25 + 35 + 45 = 105\text{ kW} > 100\text{ kW}$** (‘5 kW‘ firm deficit), and at $30\,^\circ\text{C}$ degraded loop (72 kW), chilled-water load is **$1 + 35 + 43.8 = 79.8\text{ kW} > 72\text{ kW}$** (‘7.8 kW‘ thermal deficit). Stepping down **G5 (‘45 kW‘ $\rightarrow$ ‘8 kW‘)** saves ‘37 kW‘ power and ‘36.3 kW‘ water cooling, restoring positive margin in both cases while protecting G4 and G3.

4. What Would Change If the Vessel Platform Were Modified? (Mandatory Closing Analysis)
Que cambiaria en el barco / What would change if the vessel had:
1. **A second free mast position (‘mast_4‘, $\le 60\text{ kg}$, $\le 2\text{ kW}$ air) or non-interfering roof sponson**: Both mass-feasible 4-guest sets ‘{G1, G2, G4, G5}‘ (‘7,560 kg‘) and ‘{G1, G3, G4, G5}‘ (‘9,560 kg‘) would resolve their mast/roof collision (‘G1‘ on ‘mast_3‘ at ‘1.5 kW‘, ‘G5_MAST_ANTENNA‘ on ‘mast_4‘ at ‘1.2 kW‘) without violating ‘G4‘ flight-deck/hangar clearance.
2. **A $+50\text{ kg}$ mass margin increase (‘10,050 kg‘ instead of ‘10,000 kg‘)**: ‘{G1, G2, G3, G4}‘ (‘10,035 kg‘, ‘10,323 kg-eq‘) would transition from ‘YES_IF‘ to **‘YES‘**, as its subcomponents already fit physically (‘G1‘ in ‘mast_3‘, ‘G2‘ in ‘bay2‘, ‘G3‘ in ‘bay1‘, ‘G4‘ in ‘hangar‘).
3. **An upgraded chilled-water heat exchanger (‘+20 kW‘ at $30\,^\circ\text{C}$ degraded $\rightarrow 92\text{ kW}$)**: The vessel could sustain ‘G3 (1 kW) + G4 (35 kW) + G5 (43.8 kW water) = 79.8 kW‘ during $30\,^\circ\text{C}$ loop degradation without stepping down G5 until G3 initiates its ‘25 kW‘ recharge.